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IoT Based Smart Helmet and Accident Identification System

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***Abstract*—A motorcycle frequently called motorbike or two-wheelers, which is the most used than another form of automobiles because of its low price. But another side, this is the most unsafe automobile. The accident can happen for driving fast or drunk driving. Safety and security in vehicle traveling are a pre-eminent concern for all. With the rapid urbanization and staggering growth of transport networks like two-wheeler vehicles, safety on the roads and security on the bike has emerged as an inescapable priority for us. It has expanded the rate of accidents, which leads to several damages with loss of lives. In many circumstances, we cannot able to detect the accident's location. A helmet is a form of protecting gear worn to keep safe the head from injuries. More specifically, the helmet aids the skull in protecting the brain. A smart helmet can detect the accident's locations also save lives and makes two-wheeler driving safer from previously. This paper propounds a smart helmet system to avoid the accident. The system divides into three parts helmet circuit, automobile circuit, and mobile application. At first, the helmet circuit has IR and alcohol detection sensor. The automobile circuit has a 3-axis accelerometer, Bluetooth module, relay, and load sensor. The helmet circuit sends a signal to the automobile circuit to start if the helmet is wearied and no alcohol detects. Then the automobile circuit checks the status of the load to start. 3-axis accelerometer senses crash or hit. After detecting an accident mobile application sends the accident location automatically to police and emergency contact number via the database.**

Keywords— smart helmet, accident detection, Arduino, IoT, database

I. INTRODUCTION

Per vehicle mile traveled, motorcyclists are about 30 times more likely than a passenger in a car to die in a crash. And, more than half of motorcycle fatalities in 2013 were unhelmeted riders [1]. Most motorcycle accidents that result in death are caused by a head injury. Saving a life and prevent a brain injury rider should wear a helmet. Serious injuries are greatly reduced both in severity and in frequency by the simple act of wearing a helmet.

Now accidents on roads have become a serious concern for all the thousands of people especially youngsters are either getting killed or injured every year. Transport Ministry road accident report 2014 which stresses the need to tackle this issue as soon as possible seventy-five thousand youngsters killed in road crashes last year. This clearly suggests that Indian roads present a depressing scenario most of these people who lost their lives were aged between 15 and 34 years according to the road accidents report 2014. Besides, over 82% of these victims were males. The report reveals that the age group of 15 to 34 years accounted for 53.8% of the total road accident [2]. Fatalities followed by the age group of 35 to 64 years accounted for 35.7%. About 1.2 million Indians have killed car accidents over the past decade on average one every four minutes while 5.5 million was seriously injured. The latest estimate by the World Health Organization also shows that globally world road traffic injuries are the number one cause of death among young people aged 15 to 29 years. Annually, about 3.4 lakh youngsters in this age group died in accidents according to the report prepared by the ministries transport research wing. The total number of road crashes has increased marginally from 3.86 lakhs in 2013 to 4.50 lakhs in 2014. The number of fatalities has also gone up by about 1.5 percent [3]. This shows that there is a need to pay attention to make young people more aware of road safety issues. It is a mammoth loss and calculating the social cost would reveal how big it is a rise in the number of accidents and the number of people getting killed or injured in those accidents is certainly a major concern. However, it is the responsibility of every individual especially the youngsters who come onto the roads that they have to be safe and follow the traffic rules.

Motorcycle helmets have two main protective components. First is outer shell typically made from plastic, fiberglass, or Kevlar, and second is inner, usually made of polystyrene foam. The purpose of the hard outer shell is to prevent penetration of the helmet or head and to provide structure to the soft inner liner to keep it stable upon impact. The purpose of the foam liner is to crush during impact to increase the distance and the amount of time head has to stop and to reduce its deceleration. Helmeted bikers experienced significantly fewer and less severe head and neck injuries than unhelmeted riders. People who were injured had head and neck caused by the safety helmets, each of these injuries was relatively minor, and helmets protected them from

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potentially fatal injuries. Therefore, helmets greatly reduce the number of deaths and injuries by motorcycle accidents.

The main contributions of this paper can be brief as follows: a) preventing accident, b) reducing accident causality, c) identifying the accident, and d) developing the accident-related traffic management system.

In this paper, we propose a model of a smart helmet for accident identification and preventing the accident. There use IR sensors, gas sensors, and load sensors for preventing an accident. The 3-axis accelerometer is used for detecting any accident. Arduino is used for processing sensors data and building a communication system between sensors and mobile applications. The mobile application is connected with a central monitoring system and authority can monitor every user accident history. When any accidents occur, the accident location sends to the monitoring database, and the monitoring system sends the location to the nearest hospital and police center. In Fig. 1 the main theme of this paper is described in the block diagram.

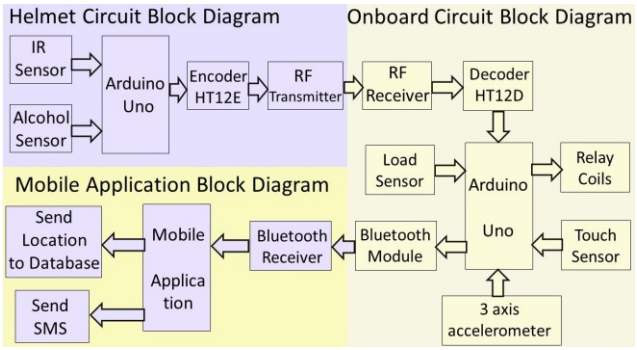


Fig. 1. Block diagram of the proposed system

Related work has been described in section II of this paper with the advantages and disadvantages. Section III there presents the methodology behind the smart helmet and accident detection system, and also described various system components' working principles. Results and discussions are providing in Section IV. In the end, the conclusion is given in Section V.

II. Related Work

There is a huge amount of work in progress in the implementation of add-on interaction systems for motorcycles. In [4] the author proposed a system where uses two sets of the control system. Those are the rider to motorcycles and motorcycle to the operation control system. All the communication happened wirelessly via Bluetooth communication from helmet to motorcycle control device. When the helmet detects any alcohol present in the air, the motorcycle will not start. However, this structure does not guarantee that the rider wears a helmet. A cost-effective system is proposed by the author in [5]. It is a ZigBee-based wireless communication system that creates a wireless network system with various sensors like a gas sensor and a speed sensor. This system notified to the rider for over speed and also alcohol present in breath. ZigBee wireless technology has a high initial cost and lower coverage area.

In [6], the author proposed a system for bike accident protection. The system aims to reduce the damage to life. The author uses an IR sensor to confirm that the bike rider wears the smart helmet. When accidents happen, the system detects the accident and sends an accident location to the nearest hospital by the GSM module. GPS tracking system detects accurate location information. For detecting an accident, there are used 3 axis accelerometer sensor.

A vision-based automatic accident detection system has been developed in [7]. The author uses a high definition camera for visual input. The raw image is processed in raspberry pi which detects the accident. A report of the accident has been notified in the database. So, authority and ambulance know where the accident happened. The system helps the authority to determine the cause of accidents.

The main cause of death in most accidents that medical services do not arrive on time. In remote places, where the accidents happen, no one can be informed to the ambulance. In [8] the author proposes a cloud base accident monitoring system. Three main units are motorcycle units, ambulance units, medical center units. The main purpose of the system to send accident locations to the nearest medical center. There is no pre-accident protection system.

III. Methodology

A. Load Sensor

Strain-Gauge type load sensors are commonly used to measure loads. Strain-Gauge is a transducer that converts force into an electrical signal. When force applies to strain gauge, it produces a voltage difference. The voltage difference will be positive or negative which defined by compressing or stretching a strain gauge. Strain gauge makes it into a flexible surface. Therefore stretching a strain gauge makes it longer and thinner, resulting to increase its internal resistance and compressing which makes it shorter and thicker, resulting to decrease its internal resistance. Changing internal resistance means the voltage difference.

The S-type load cell used in this paper for detecting overload. There is a threshold load value for safety driving. When the sensor detects any overload, the rider cannot start the motorcycle. The threshold load value is about 180 kg which is the weight of two adult people. The S-type load cell is situated in the spring of the motorcycle.

B. MQ-3 Gas Sensor

MQ-3 module is the mostly use electrochemical gas sensor to detect alcohol. Tin Dioxide (SnO2) is used in the sensor which conductivity is lower than clean air. As the alcohol concentration is high, the conductivity of the sensor will be high. Therefore, the internal resistance of the sensor goes lower. The resistance of two points varies for detecting the presence of alcohol in the rider's breath.

C. Database

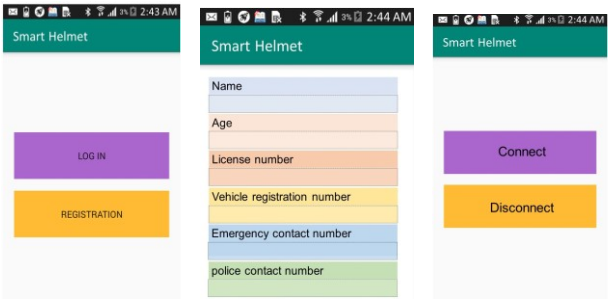
The database supports a large number of user data using the mobile application. Firebase is an open and free database from google which processes real-time data that can be stored or monetized. When any motorcycle rider register to the mobile application, all the data is stored in the database which can be monetized and also categorized according to the operator's preference, remotely and real-time. When an accident happens, the location of the accident place and motorcycle rider's information sent to the nearest hospital and emergency contact numbers. The operator can also be categorized all the accident data with time and location. Rider's other information like over speed data and overload

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data are stored in the database. Law enforcement authorities can monitor individual users or groups of users. This will help law enforcement authority and rider for safe traffic.

D. The User Interface

The mobile application is a graphical user interface (GUI) which is designed to interact between human and machine. GUI is designed into the android studio with the help of JAVA programming language. At first in the mobile application, the user registers his/her name along with emergency contact number and other mandatory information. After registration, there is a login option for login. Then there is a connect option for connecting to the device by Bluetooth communication. When rider drives at over speed, a notification window pop-up in the mobile application. It also happens in overload conditions.



a b c

Fig. 2. a) mobile application main window. b) user registration window. c) connect to onboard circuit window

E. Accelerometers

An accelerometer is an electromechanical device used to measure acceleration. Acceleration means the rate of change of velocity. Therefore accelerometer measures the acceleration of force like gravity. The basic physics of acceleration measurement using a spring and mass system. A body of mass m is attached to a wall by a spring having spring coefficient k. When force F is applied to mass m, it is displaced in the direction of the force by amount x with acceleration a. Using Newton law, and writing the force conserving equation, we get force due to spring tension.

$$F = ma = Fs \qquad (1)$$

The force due to spring is written as kx where x is the displacement of the body from the initial rest position. Equating the forces we get

$$ma=kx \qquad (2)$$

Thus, we see that acceleration is a function of displacement.

$$A = f(x) \qquad (3)$$

If we can measure displacement somehow, we can calculate the acceleration on the body. There are different methods to measure displacement 'x'. In the capacitive technique, we know from basic physics that

$$C=\varepsilon*A/d \qquad (4)$$

Where d is the displacement between the two plates forming the capacitance. By varying the distance between the plates, we can change the capacitance 'C' which measured electronically by using a signal conditioning circuit. The capacitors have two plates. One is the movable plate attached to the mass m. The other is fixed at the initial position. When the body accelerates due to force 'f', it causes a change in the displacement 'x' which is dependent on the acceleration 'a' of the body. Due to 'x', the capacitance across the fixed and movable plates changes. By measuring this C, we can calculate the value of 'x' which helps us to infer the value of 'a'. This is the principle of measuring acceleration.

However one cannot employ such big spring-mass systems inside accelerometer IC's. This is where MEMS (micro-electromechanical system) comes into play. These systems contain both mechanical and electronic components which fabricated at the scale of a micrometer and employed inside the accelerometer IC. There is an internal movable assembly having a small mass and connected to outer assembly using spring contacts. The movable assembly also has plates that form a capacitor with plates of outer fixed assembly. As the system moves due to acceleration, the internal assembly moves and due to displacement, the value of different capacitors changes. By measuring these changes in capacitance, we can infer the acceleration acting on the body. Having such multiple MEMS systems in different planes X, Y, Z the accelerometer gives the readings of acceleration in different directions as seen by the body.

In the earth's surface, the gravitational force is 1g (9.81 m/s2). 1g is the reference gravitational force. When accelerometer moving the gravitational force is above 1g. As the speed increases, the gravitational force will be increased. The deviation from 1g to the resultant force, the speed is measured from the deviation. But accelerometer shows about zero gravitational force during fall. When acceleration suddenly stops, the resultant force in much higher than the normal acceleration force. In accident condition gravitational force some time very high, sometimes near to zero. The time interval is very short. Variation of gravitational force in a short period indicates an accident. There uses a 3 axis capacitive type accelerometer that has a great operation range and available in a wide variety of temperature ranges.

F. Working Principal

Every year, motorcycle-related accidents lead to catastrophic deaths, cost enormous money in several nations. A smart helmet helps to reduce those accidents casualty and saves lives. In this prototype design, the smart helmet has two sets of circuits. One is the helmet circuit, other is the onboard circuit and also has a mobile application.

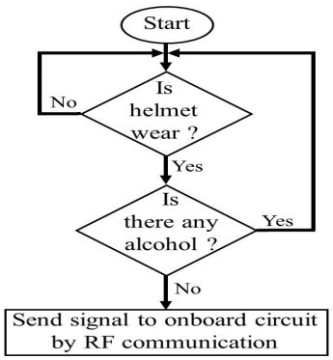


Fig. 3. Working principle of helmet circuit

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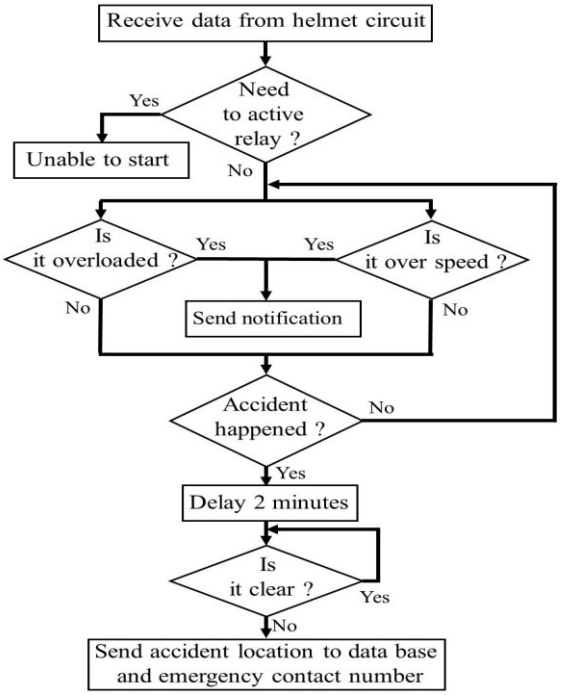


Fig. 4. Working principle of the onboard circuit

The smart helmet circuit consists of two sensors, an alcohol sensor, and an IR sensor. Alcohol sensor measures alcohol present in the rider's breath and the IR sensor detects the presence of the rider's head. If there any alcohol presents or doesn't wear the helmet, an abnormality signal sent to the onboard circuit by RF signal at the frequency of 433 MHz Any abnormality found in the helmet circuit which gives a signal to the microcontroller for high pulse output to the encoder IC HT12E. The encoder IC sent the signal at RF frequency to the decoder antenna. When the antenna receives any signal, decoder IC HT12D decodes the signal and sends it to Arduino Uno for the further process. If there is any abnormality in the helmet circuit, the relay will activate to disconnect the battery circuit. Therefore, the motorcycle can't start. Before starting the motorcycle, the rider login to the mobile application. After starting the load sensor measures the current load of the rider. If there is any excess load for two minutes, the rider will notify through the mobile application. For over speed biker also notify. The over speed and overload data sent to the database by the mobile application. The over speed and overload data send to the mobile application through Bluetooth communication. There choose two kinds of different communication (Bluetooth and radio frequency communication) for avoiding interference.

If there is an acceleration variation in a short period, Arduino detects it as an accident. When an accident occurs, Arduino waits two minutes for a touch sensor response. Touch sensor used for the mild accident. When an accident occurs, if the rider touches the touch sensor, that accident report as a mild accident that can't send any message to the emergency contact number. If the rider can't touch the sensor, accidents' current location sends immediately to the database and emergency contact number automatically. The mobile application detects the current location by the mobile's built-in GPS (Global Positioning System) module. And GSM (Global System for Mobile communication) sends the location to the emergency contact number. The accident location is sent to the database via the internet. When the database operator notified any accidents, the operator's system sends the location and other details to the nearest hospital and law enforcement agencies for immediate action.

IV. Results and Discussion

The smart helmet ensures motorcycle rider safety and also takes the necessary steps to reduce accident causality. if any safety features are violated, the system can't permit the rider for starting the motorcycle. The alcohol sensor's accuracy depends on the distance between the mouth and sensor. When the distance above 15 cm, the sensor read <150 mg/L means rider has low blood alcohol content (BAC). The alcohol level reaches >350 mg/L means rider has higher BAC, as the distance reduces to less than 6 cm. In this system, there use a smartphone built-in a GPS. Low mobile signal, high-speed motion, older version GPS module are the main cause for reducing GPS accuracy. In this system, the accuracy of the GPS module nearly 1 to 5 meters. On the other hand, the accident detection sensor, the 3 axis accelerometer sensor has a high accuracy rate. After calibration, the accelerometer's value changes nearly 1.280g from the original value. It has 100Hz bandwidth for detecting accidents. In the testing period, accident detection accuracy is nearly one hundred percent.

V. Conclusion

The proposed system aims to prevent motorcycle accidents and also reduce the fatality of accidents. Family members, law enforcement agencies, and the nearest hospital are informed about accidents and also accidents location. Accidents detection algorithm has high accuracy and automatically record, detect, and report the accidents mediately. The driving data helps the rider to improve driving. This system makes a habit to wear a helmet among motorcycle riders. The smart helmet would make a motorcycle journey more protected and safer.

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